Trial SET #4 (math) :



- Don't use calculator and any other materials, including formulae and tables;
- You have only 60 minutes to complete the test;
- Try to do your best and Good Luck!!!

You should be familiar with:

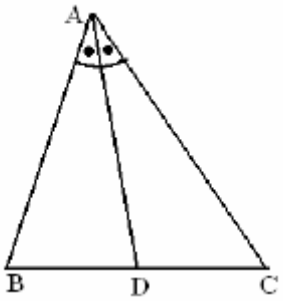
- word problems;
- angles in triangles;
- number properties;
- combinatorics;
- integrals;
- trigonometry;
- arithmetic sequences;
- roots;
- areas in quadrilaterals;
- circles;
- logarithms;
- derivatives;

Name of the student _____

1. A cooler filled with cans of soda weighs 20 pounds. After three-quarters of the cans of soda have been consumed, the cooler, together with the remaining sodas, weighs 14 pounds. If each can of soda weighs the same, how much does the empty cooler weigh?

A. 10; B. 12; C. 14; D. 9; E. 8;

2. In the given figure AD is angle bisector, $\angle B - \angle C = 20^o$ then find $\angle BDA$.



A. 55; B. 65; C. 75; D. 80; E. 85;

3. $\left(\sqrt{10}\right)^{2-\lg 4} = ?$

A) 5. B) 4. C) 8. D) $2\sqrt{10}$ E) $4\sqrt{5}$.

	FIRST PLACE	SECOND PLACE	THIRD PLACE
	(8 points)	(4 points)	(2 points)
EVENT ①	TEAM A	TEAM B	TEAM C
EVENT ②	TEAM B	TEAM A	TEAM C

4. The results of two games involving 3 teams are shown above. Thus, we have the following standings: A and B both have 12 points, and C has 4 points. Assuming no ties, what is the least number of additional games that Team C will have to play in order to have the highest total score?

A. 1; B. 2; C. 3; D. 4; E. Third team will never win;

5. An Ameba is an organic life form that divides into two Amebas each round hour. If at a certain round hour, two Amebas were placed in a jar, how many Amebas will be in the jar in N hours?

A. 2N; B. 2^{2N}; C. 2^{N+1}; D. 2^{N-1}; E. 2^{N};

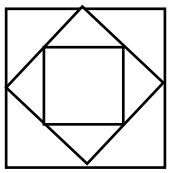
6. $\dfrac{\sqrt[3]{2}\cdot\sqrt[4]{8}\cdot\sqrt[6]{4}}{2\cdot\sqrt[12]{2}} = ?$

A. $\sqrt[3]{2}$; B. $\sqrt[12]{2^5}$; C. $\sqrt[4]{2^3}$; D. 2; E. 1;

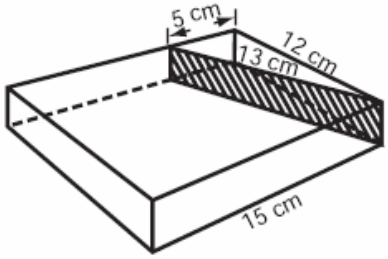
7. An investment yielded an interest payment of $350 each month when the annual interest rate was 9%, what is the amount of the investment that should be invested with an annual interest rate of 8% if We want to gain 15% more each month?

A. $60,375; B. $50,400; C. $41,300; D. $32,500; E. $25,100;

8. In the figure a square is drawn connecting the middle points of each side of a square with a side length 6 and this procedure is repeated. What is the perimeter of the 9th square that will be obtained? (Assume that the square with side 6 is the first).



A. $\dfrac{3}{2}$ B. $\dfrac{3\sqrt{2}}{2}$ C. 3 D. $3\sqrt{2}$ E. $6\sqrt{2}$



9. The rectangular box above has a rectangular dividing wall inside, as shown. The dividing wall has an area of 39 cm^2. What is the volume of the larger compartment?

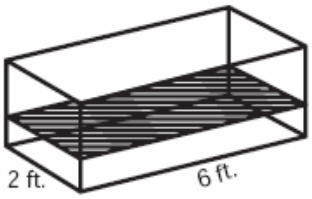
A. 90 cm^3; B. 180 cm^3; C. 360 cm^3; D. 450 cm^3; E. 540 cm^3;

10. A computer is programmed to add 3 to the number N, multiply the result by 3, subtract 3, and divide this result by 3. The computer answer will be

A. $N + 1$; B. $N + 2$; C. N; D. $N - 2$; E. $N + \frac{1}{3}$;

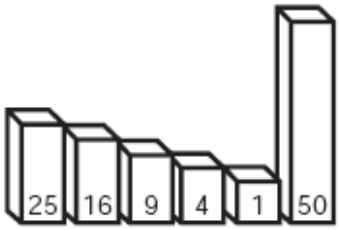
11. Meg and Bob are among the 5 participants in a cycling race. If each participant finishes the race and no two participants finish at the same time, in how many different possible orders can the participants finish the race so that Meg finishes ahead of Bob?

A. 24
B. 30
C. 60
D. 90
E. 120



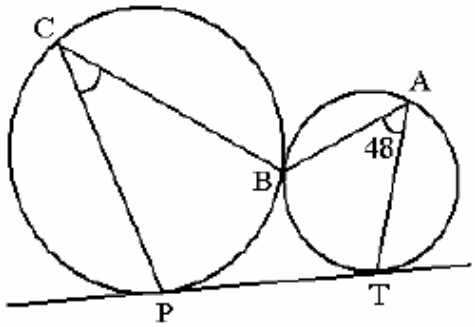
12. The figure above shows water in a tank whose base is 2 feet by 6 feet. If a rectangular solid whose dimensions are 1 foot by 1 foot by 2 feet is totally immersed in the water, how many *inches* will the water rise? (1 foot = 12 inches)

A. 1/6; B. 1; C. 2; D. 3; E. 12;



13. Six containers, whose capacities in cubic centimeters are shown, appear in the figure above. The 25-cubic-centimeter container is filled with flour, and the rest are empty. The contents of the 25-cubic-centimeter container are used to fill the 16-cubic centimeter container, and the excess is dumped into the 50-cubic-centimeter container. Then the 16-cubic-centimeter container is used to fill the 9-cubic-centi meter container, and the excess is dumped into the 50-cubiccentimeter container. The process is repeated until all containers, except the 1-cubic-centi meter and the 50-cubiccentimeter containers, are empty. What percent of the 50-cubic-centimeter container is *empty*?

A. 24%; B. 48%; C. 50%; D. 52%; E. 76%;

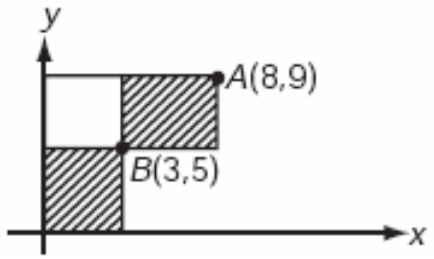


14. In the given figure P, T and B are tangent points. If $m\angle BAT = 48^0$ then find $m\angle PCB$.

A) 32 B) 38 C) 42 D) 44 E) 48

15. A number was divided by 5 and the result was increased by 7. Then it was multiplied by 5. Comparing with the number what is the change in result?

A. it is greater by 7; B. it is less by 7; C. it is greater by 35;
D. it is the same; E. it is 7 times as large;



16. Points *A* and *B* have coordinates as shown in the figure above. Find the combined area of the two shaded rectangles.

A. 20; B. 26; C. 32; D. 35; E. 87;

17. If an integer n is divisible by both 18 and 24, then it must also be divisible by

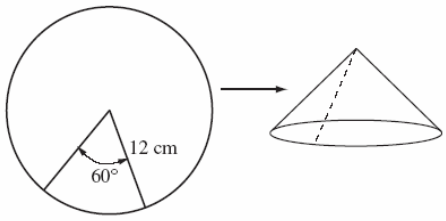
A. 12; B. 16; C. 27; D. 120.; E. 432;

18. If the sum of 5 consecutive positive integers is w, in terms of w, which of the following represents the sum of the next 5 consecutive positive integers?

A. $w + 5$; B. $5w + 5$; C. $5w + 25$; D. $w + 25$; E. $w^2 + 25$;

19. The integer part of a positive number is the part before the decimal point; The decimal part of a positive number is the part after the decimal point. For example the integer part of 4.32 is '4' and the decimal part is '0.32'. Which of the following numbers has a decimal part five times less than its integer part?

A. 2.5; B. 4.8; C. 2.45; D. 3.40; E. 2.42;



20. A 60° wedge is cut from a paper circle with radius 12 centimeters and the paper is then folded into a cone, as shown in the figure above. What is the height, in centimeters, of this cone?

A. $\sqrt{10\pi}$; B. $\sqrt{44}$; C. $\sqrt{70}$; D. $\sqrt{90}$; E. $\sqrt{140}$;

21. Find the derivative of f(x) = $(\sin\frac{\pi}{2}-2x)^3$.

A) $3(\sin\frac{\pi}{2}-2x)^2$; B) –2x; C) $-6(1-2x)^2$; D) $(1-2x)^3$; E) cos2x;

22. If C is a constant, what is the value of $\int(3x+2)^4dx$?

A. $\frac{(3x+2)^4}{4}+C$; B. $12(3x+2)^3+C$; C. $\frac{3}{5}(2x+3)^5+C$; D. $\frac{(3x+2)^5}{15}+C$;

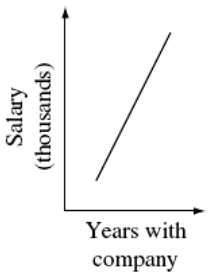
E. $3\cdot\frac{(3x+2)^5}{16}+C$;

SALARY GROWTH AT ACME PLUS CO.

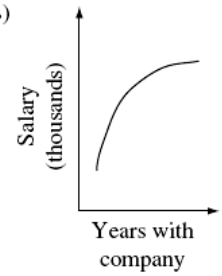
YEARS WITH COMPANY	1	2	3	4	5
SALARY (in thousands)	32	33	36	42	51

23. Which of the following graphs best represents the information in the table above?

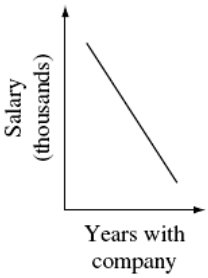
(A)



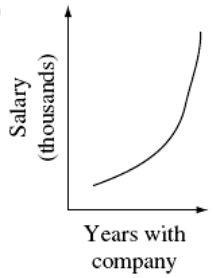
(B)



(C)



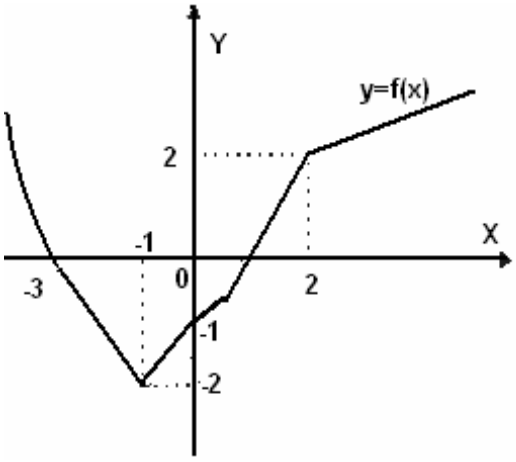
(D)



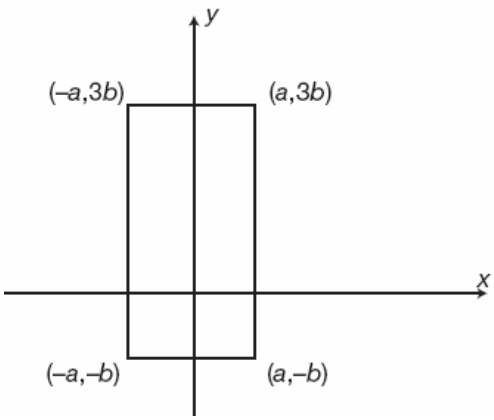
(E)



24. In the given figure, the graph of the function $y = f(x)$ is given. What is $f\left(f\left(0\right)\right)$ equal to?



A) – 3 B) – 2 C) – 1 D) 0 E) 2



25. In the diagram above, what is the area of the rectangle?

A. $6ab$ square units; B. $8ab$ square units; C. $9b^2$ square units;
D. $12ab$ square units; E. $16b$ square units;

26. Find the derivative of 2^{-3x}.

A) $-3 \cdot 2^{-3x} \cdot \ln 2$; B) $-3x.2^{-3x} \cdot \ln 2$; C) $2^{-3x} \cdot \ln 2$;
D) $-3 \cdot 2^{-3x}$; E) $-2x \cdot 2^{-3x} \cdot \ln 2$;

27. If $ad + \frac{2}{ad} = 4$, then what is $3a^2d^2 + \frac{12}{a^2d^2} = ?$

A. 16; B. 26; C. 36; D. 46; E. 56;

28. Which one of the following propositions is false?
A) $(x^2=4) \Rightarrow (x = -2)$ or $(x = 2)$
B) $(x < 5) \Rightarrow (3x-4 < 11)$
C) $(2x+1 \neq 5) \Rightarrow (x \neq 2)$
D) $(xy = 0) \Rightarrow (xyz = 0)$
E) $(x^4 = 9) \Rightarrow (x^2 = -3)$ and $(x^2 = 3)$

29. Simplify: $\frac{\cos x}{1+tgx}-\frac{\sin x}{1+ctgx}$

A) $\sin x+\cos x$ B) $\frac{2}{\cos x}$ C) 0 D) $2\sin x$ E) $\cos x-\sin x$

30. $\int_{\frac{\pi}{2}}^{\pi}(\sin\frac{x}{2}+\cos\frac{x}{2})^2\,dx = ?$

A) π B) $\frac{\pi}{2}$ C) $\frac{1}{4}$ D) $\frac{\pi}{2}+1$ E) $\frac{\pi-2}{2}$